

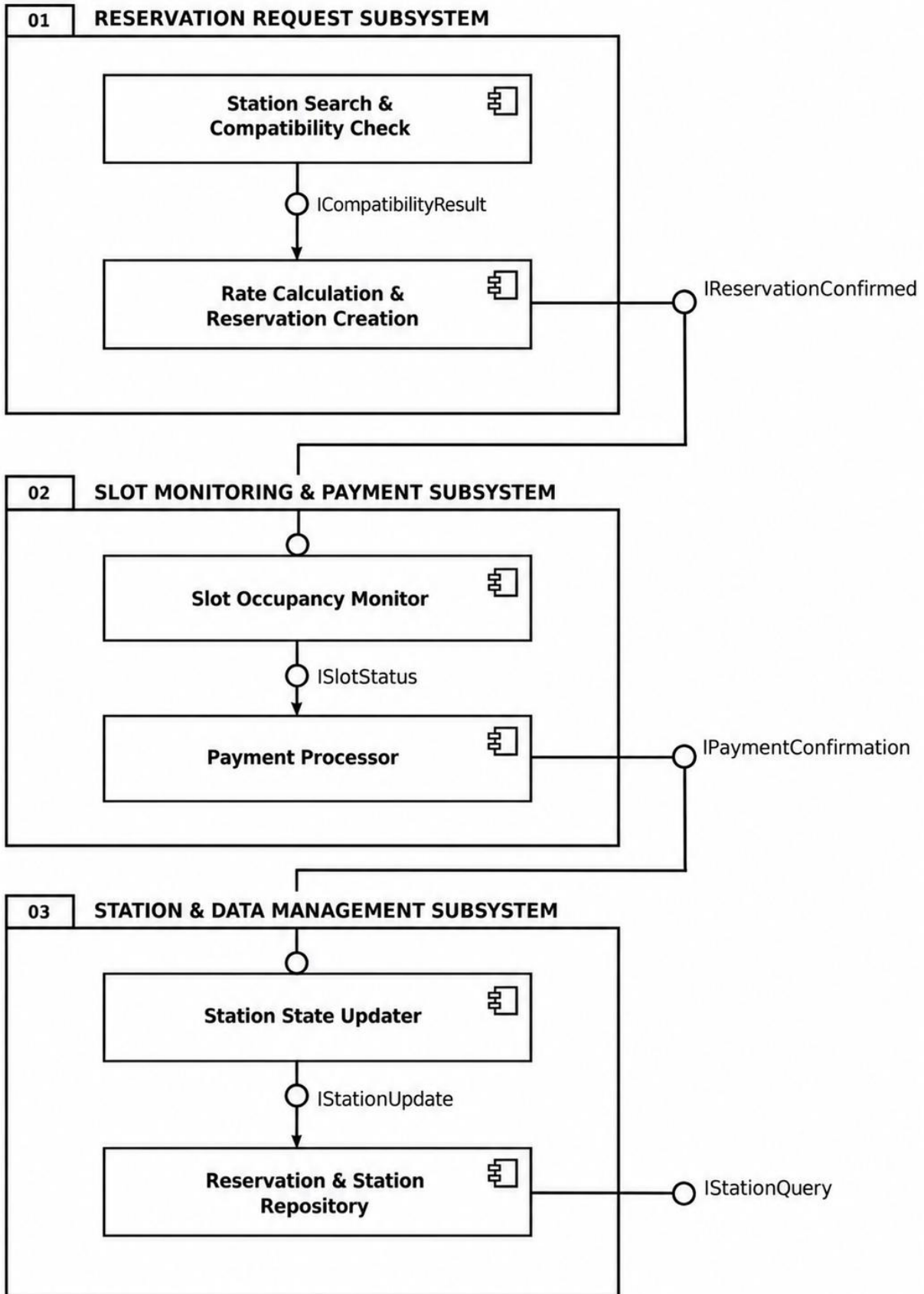
SOFTWARE ENGINEERING

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Architecture Selection:

Layered Architecture

We chose the Layered Architecture for the EV Charging Station Reservation System. The first reason for selecting this architecture is clear separation of concerns. The system performs several distinct tasks, including presenting driver- and operator-facing interfaces, coordinating reservations, checking charger compatibility, calculating rates, monitoring slot occupancy, sending notifications, processing payments, and persisting reservation and station data. By separating these responsibilities into the Presentation, Business/Application, Data Access, and Database layers, the system becomes easier to understand, maintain, test, and extend.

The second reason is that the system requires a structured and secure request-processing flow. In the proposed architecture, an EV driver's request first enters through the EV Driver Interface in the Presentation Layer and is handed to the Reservation Management component in the Business/Application Layer. Reservation Management coordinates the Charger Compatibility Service and Rate Calculation Service to validate the request and quote a price, then creates the reservation. Once a slot is reserved, the Slot Monitoring Service tracks occupancy and automatically releases the socket with a penalty if the driver does not plug in within ten minutes, while the Notification Service keeps the driver informed. Payment is delegated to the external Payment Service, and all reservation and station state is written through the Reservation Repository and Station Repository in the Data Access Layer down to the EV Charging Database. This clear, layered flow ensures that each stage performs only its own responsibility.

A major security advantage of the selected architecture is that payment handling is isolated behind the external Payment Service interface rather than being embedded in the business logic, so card and transaction details never need to pass through or be stored by the Reservation Management component. Similarly, routing all persistence through the Reservation Repository and Station Repository means the Business/Application Layer never talks to the EV Charging Database directly, which keeps access to user, vehicle, and pricing data centrally controlled and auditable.

A key performance benefit is that separating presentation, business logic, data access, and storage into distinct layers allows each part of the system to be scaled and optimized independently. In particular, the Reservation Repository and Station Repository can be optimized for high-frequency reads and writes during peak charging hours, the Slot Monitoring Service can run its ten-minute timers independently of the request path, and the Notification Service can process alerts asynchronously without blocking new reservations. Therefore, the Layered Architecture provides a clear, modular, secure, and maintainable solution for the EV Charging Station Reservation System. It supports the required separation between driver interaction, business logic, data access, and storage while addressing the important security and performance requirements of the system.